

Venkanna Babu Guthula

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SUMMARY

I am a Ph.D. student specialising in computer vision for Earth observation, with three years of industry and research experience in remote sensing image analysis and machine learning. I am passionate about leveraging artificial intelligence and high-performance computing to extract meaningful insights from all kinds of imagery.

EDUCATION

Ph.D. in Computer Vision for Earth Observation 02/2023-Present
Department of Computer Science, University of Copenhagen, Denmark

- Conducting research on malaria risk assessment in African cities using remote sensing and machine learning
- Developing computer vision methods using Convolutional Neural Networks and Vision Transformers for semantic and instance segmentation
- Focusing on methods for segmenting individual buildings, classifying their attributes, and aligning different image modalities

M.Sc. in Geo-Information Science & Earth Observation 09/2018-06/2020
Faculty of Geo-Information Science and Earth Observation, University of Twente, the Netherlands

- **Thesis:** Automatic road extraction from high-resolution remote sensing imagery using fully convolutional networks and transfer learning
- Studied basic concepts in machine learning and remote sensing
- Started exploring machine learning frameworks such as Scikit-learn, Pytorch and TensorFlow
- Gained hands-on experience in geospatial data analysis using Python, JavaScript, R, and GIS software (QGIS, ArcGIS, and SNAP)

B.Tech. in Geo-Informatics 09/2014-03/2017
Andhra University, Visakhapatnam, India

- **Thesis:** Web-GIS-based comparative analysis of rainfall in the Thatipudi Reservoir catchment area and its impact on reservoir capacity
- Studied techniques for estimating rainfall and water surface area using remote sensing imagery
- Developed a web-based application for geospatial data visualization

EXPERIENCE

Scientist I - EO 12/2021-01/2023
Cropln Technology Solutions, Bangalore, India.

- Developed a scalable model for cloud and shadow detection in Sentinel-2 imagery using classical machine learning algorithms
- Utilized AWS services for data analysis and machine learning model development
- Applied deep learning techniques for land boundary delineation from high-resolution satellite imagery

GIS-Python Developer

09/2020-12/2021

Wildlife Institute of India, Dehradun, India

- Developed deep learning models for species classification on top of MegaDetector to classify Indian mammals (<https://github.com/bhlab/SpSeg>)
- Maintained and contributed to *Linkage Mapper*, a tool for mapping wildlife corridors (<https://github.com/linkagescape/linkage-mapper>)
- Performed remote sensing image classification using machine learning algorithms in Google Earth Engine

Map-Technician

09/2017-09/2018

Mapbox Technologies, Bangalore, India

- Researched and designed workflows to improve transportation assets for enhanced navigation experiences
- Planned and organized field mapping events for large-scale geospatial data collection
- Added, validated, and enhanced base data in OpenStreetMap using GPS, remote sensing imagery, and street-level imagery

ACHIEVEMENTS

2020 Awarded the Golden Jubilee Scholarship at the Indian Institute of Remote Sensing for outstanding performance in M.Sc.

2019 Received a scholarship to attend and present a lightning talk at the *State of the Map Asia 2019* Conference, Dhaka, Bangladesh

LANGUAGES

Telugu, English and Hindi

SELECTED JOURNAL PAPERS

Brandt, M., Gominski, D., Reiner, F., Kariryaa, A., **Guthula**, V. B., Ciais, P., Tong, X., Zhang, W., Govindarajulu, D., Ortiz-Gonzalo, D., et al. (2024). Severe decline in large farmland trees in India over the past decade. *Nature Sustainability*, 1–9.

Guthula, V. B., Oehmcke, S., Chilaule, R., Zhang, H., Lang, N., Kariryaa, A., Mottelson, J., & Igel, C. (2025). Drone imagery for roof detection, classification, and segmentation to support mosquito-borne disease risk assessment: The nacala-roof-material dataset. *Science of Remote Sensing*, 12, 100306.

Guthula, V. B., Shrotriya, S., Nigam, P., Goyal, S. P., Mohan, D., & Habib, B. (2022). Biodiversity significance of small habitat patches: More than half of Indian bird species are in academic campuses. *Landscape and Urban Planning*, 228, 104552.

SELECTED CONFERENCE PAPERS

Guthula, V. B., Kehlet, J. J. Å., Kariryaa, A., Lang, N., Oehmcke, S., & Igel, C. (2025). Beyond building footprints: Probing dinov3 to map roof material and geometry. *1st Workshop on Advances in Representation Learning for Earth Observation (REO @ EurIPS 2025)*.

Guthula, V. B., Krause, O., Gominski, D., Zhang, H., Mottelson, J., Kariryaa, A., Lang, N., & Igel, C. (2026). Align and segment: Unsupervised learning for building segmentation from misaligned labels. *European Conference on Computer Vision (ECCV)*.